Astro 410 Project Outline: Parameterization of Dark Matter Halos using the Markov Chain Monte Carlo Algorithm

Leo Kajiwara

October 2026

1 Purpose

Galaxy rotation curves give orbital speed as a function of the radius from a galaxies center. By comparing the observed speeds with the speeds predicted by the visible matter, a noticeable discrepancy is observed. A rotation curve described by the visible matter would decay as the radius reaches larger distances, but the observed rotation curves of galaxies has a flatter profile. This discrepancy is caused by dark-matter halos.

This project builds models of rotation curves by taking account of the dark-matter halos along with the visible matter contributions. It uses a Markov chain Monte Carlo (MCMC) algorithm to estimate plausible halo parameters such as core density, core radius, etc. There are a few dark matter halo profiles that can describe dark matter halos, but the one I plan to use is the Burkert Ansatz Profile, which represents a halo with a central density core. The goal is to learn whether the two halo parameters can reproduce accurate models for any given rotation curve.

The data base that I am using to collect observed rotation curves and visible matter contributions to the rotation curves is SPARC from the Lelli et al. [2016] paper.

2 Data and Model

The data inputs that this algorithm takes is radius, observed velocity at each radius, the speed uncertainty, and the visible matter contributions to the rotation speed. The model for the Burkert Ansatz model has two parameters: central density and the core radius.

With manual inputs for the two parameters, I have been able to successfully model rotation curves by combining the visible matter contributions and the Burkert Ansatz Dark Matter contributions. For each proposed pair of parameters, the MCMC algorithm integrates the density profile to find the enclosed-mass and calculates the dark-matter's contribution to the rotation curve at each

radii from the data set. Then it combines the dark-matter's contribution to the visible matter's contribution. The log-likelihood scores how well a proposed parameter pair predicts the data with the speed uncertainties taken into account. The prior states which parameter values are allowed and which aren't. MCMC proposes parameter values and accepts or rejects them depending on the posterior probability (the likelihood times the prior). The output is a chain of sampled parameter pairs. Then it plots the modeled rotation curve with the parameters the MCMC algorithm lands on.

3 Current Setback

The algorithm was taking a while to run because of the expensive mass calculations. This caused me to pivot from using the Simpson integration method to an analytic solution. There were plenty of syntax errors that caused errors.

4 Future Tests / Implementations

Different density profiles can be used to create rotation curve models with different sets of parameters to be inputted. These different profiles can be compared with each other too. However, I need to find a way to get the MCMC algorithm working properly first before I can start thinking about comparing different density profiles.

5 First Draft notes

- Burkert Ansatz Density Profile
- Two Parameters: core density, central core radius
- **Useful Paper:** Lucie-Smith et al. [2024]
- **Useful Archive of Galaxies** Lelli et al. [2016]

Current progress:

Working method in producing dark matter halo rotation curves given central core density and central core radius parameters. Archive of galaxies will be used to retrieve observed rotation curves as well as the visible matter contribution of the rotation curves. The visible matter contribution will be combined with the dark matter contribution to produce a, hopefully, accurate rotation curve.

Next steps, make MCMC algorithm to fine parameters of the halo

References

Federico Lelli, Stacy S. McGaugh, and James M. Schombert. SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves. , 152(6):157, December 2016. doi: 10.3847/0004-6256/152/6/157.

Luisa Lucie-Smith, Hiranya V. Peiris, and Andrew Pontzen. Explaining dark matter halo density profiles with neural networks. *Phys. Rev. Lett.*, 132:031001, Jan 2024. doi: 10.1103/PhysRevLett.132.031001. URL `https://link.aps.org/doi/10.1103/PhysRevLett.132.031001`.